

Non-normability of the Brezis–Seeger–Van Schaftingen–Yung spaces for $-1 < \gamma < 0$

FA/Banach run `fa_banach_001`, lane 1

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Source Metadata

- Source paper: Haim Brezis, Andreas Seeger, Jean Van Schaftingen, and Po-Lam Yung, *Families of functionals representing Sobolev norms*, arXiv:2109.02930.
- Local source PDF: `source_paper.pdf`.
- Extracted question location: Section 7.1, page 37 of the arXiv PDF.
- Claimed status: partial solution. The packet answers the normability question for $-1 < \gamma < 0$ and leaves the endpoint $\gamma = -1$ open.

Open Problem Extract

For $-1 \leq \gamma < 0$, the source paper defines

$$\|u\|_{\dot{B}V(\gamma)} := \|\nabla u\|_{\mathcal{M}} + \sup_{\lambda > 0} \lambda \nu_{\gamma}(E_{\lambda, \gamma}[u])$$

and the corresponding subspace $\dot{W}^{1,1}(\gamma)$ of $\dot{W}^{1,1}$. The paper asks:

“Are these spaces normable in the range $\gamma \in [-1, 0)$?”

7. PERSPECTIVES AND OPEN PROBLEMS

7.1. Subspaces of $\dot{W}^{1,1}$ and $\dot{B}V$ and related spaces

The failure of the upper bounds for $[Q_\gamma u]_{L^{1,\infty}(\mathbb{R}^{2N}, \nu_\gamma)}$ for $\gamma \in [-1, 0)$ raises a number of interesting questions. Consider the space $BV(\gamma)$ consisting of all $\dot{B}V$ functions satisfying

$$(7.1) \quad \|u\|_{\dot{B}V(\gamma)} := \|\nabla u\|_{\mathcal{M}} + \sup_{\lambda > 0} \lambda \nu_\gamma(E_{\lambda, \gamma}[u]) < \infty$$

and the corresponding subspace $\dot{W}^{1,1}(\gamma)$ of $\dot{W}^{1,1}$.

Embeddings. We proved in this paper that for $\gamma \notin [-1, 0]$ we have $\dot{B}V(\gamma) = \dot{B}V$ and $\dot{W}^{1,1}(\gamma) = \dot{W}^{1,1}$. It is natural to ask how in the range $-1 \leq \gamma < 0$ the proper subspaces $\dot{B}V(\gamma)$ and $\dot{W}^{1,1}(\gamma)$ relate to other families of function spaces, in particular to the Hardy-Sobolev space $\dot{F}_{1,2}^1$, another subspace of $\dot{W}^{1,1}$.

Triangle inequalities. The spaces $\dot{W}^{1,1}(\gamma)$ and $\dot{B}V(\gamma)$ are defined via $L^{1,\infty}$ -quasi-norms, and the space $L^{1,\infty}$ is not normable (unlike $L^{p,\infty}$ for $1 < p < \infty$ which is normable [13]). However Theorem 1.4 tells us that $\dot{W}^{1,1}(\gamma)$ and $\dot{B}V(\gamma)$ are normable for $\gamma \notin [-1, 0]$. Are these spaces normable in the range $\gamma \in [-1, 0)$?

Related quasi-norms. Consider for $0 < s \leq 1$

$$\|u\|_{(p,s,\gamma)} = \left[\frac{u(x) - u(y)}{|x - y|^{\frac{2}{p} + s}} \right]_{L^{p,\infty}(\mathbb{R}^{2N}, \nu_\gamma)}.$$

It is an obvious consequence of Theorem 1.3 that for $s = 1$ and fixed $p > 1$, these expressions define equivalent (semi/quasi)-norms on C_c^∞ as γ varies over $\mathbb{R} \setminus \{0\}$. It would be interesting to find a more direct proof of this observation which does not involve the relation with $\dot{W}^{1,p}$. We note that the equivalence for varying γ breaks down for $0 < s < 1$. This result, and more about the spaces for which $\|u\|_{(p,s,\gamma)} < \infty$ with $0 < s < 1$, such as their connection to Besov spaces and interpolation, can be found in [20].

7.2. Other limit functionals

Our results, combined with the various developments presented in [5, 6, 16, 18], suggest several possible directions of research.

Can one prove a generalization of (1.14), (1.16) where the supremum is replaced by the $\liminf_{\lambda \rightarrow \infty}$ when $\gamma > 0$ and by a $\liminf_{\lambda \rightarrow 0^+}$ when $\gamma < 0$. More precisely, for $1 < p < \infty$

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Definitions

For $u \in L_{\text{loc}}^1(\mathbb{R}^N)$, set

$$E_{\lambda, \gamma}[u] = \left\{ (x, y) \in \mathbb{R}^N \times \mathbb{R}^N : x \neq y, \frac{|u(x) - u(y)|}{|x - y|^{1+\gamma}} > \lambda \right\},$$

and

$$d\nu_\gamma(x, y) = |x - y|^{\gamma-N} dx dy.$$

For $-1 < \gamma < 0$, define

$$\mathfrak{s}_\gamma(u) := \sup_{\lambda > 0} \lambda \nu_\gamma(E_{\lambda, \gamma}[u]), \quad \mathfrak{q}_\gamma(u) := \|\nabla u\|_{L^1(\mathbb{R}^N)} + \mathfrak{s}_\gamma(u)$$

on $\dot{W}^{1,1}(\gamma)$.

Proof Intuition

The paper constructs Cantor-type functions u_m for which the weak-type term grows like m while the ordinary $W^{1,1}$ size stays controlled. The extra observation is that these same u_m are not exotic single atoms: they are convex averages of 2^m one-scale transition atoms. Each one-scale atom has uniformly bounded $\mathfrak{q}_\gamma$ because a single Lipschitz transition layer has uniformly bounded weak endpoint functional when $-1 < \gamma < 0$.

If the quasi-norm topology came from an equivalent norm, then the convex hull of the quasi-unit ball would have to be quasi-bounded. The functions u_m give convex averages of uniformly bounded atoms whose $\mathfrak{q}_\gamma$ -size tends to infinity. This contradicts normability.

Theorem

Theorem 1. *Let $N \geq 1$ and $-1 < \gamma < 0$. The quasi-normed spaces $\dot{W}^{1,1}(\gamma)$ and $\dot{BV}(\gamma)$ defined in Section 7.1 of Brezis–Seeger–Van Schaftingen–Yung are not normable, in the sense that their natural quasi-norm topologies are not induced by any equivalent norm.*

Proof

We first recall a standard criterion. Let (X, q) be a quasi-normed vector space. If the topology of q is induced by an equivalent norm, then the convex hull of the q -unit ball is q -bounded. Indeed, if $a\|x\| \leq q(x) \leq b\|x\|$, every convex combination of points satisfying $q(x) \leq 1$ has norm at most a^{-1} and therefore q -size at most ba^{-1} .

It is enough to contradict this criterion in $\dot{W}^{1,1}(\gamma)$; the conclusion for $\dot{BV}(\gamma)$ follows because $\dot{W}^{1,1}(\gamma)$ is a linear subspace of $\dot{BV}(\gamma)$ and the restricted quasi-norm is the same.

Let

$$\rho = 2^{-1/(1+\gamma)} \in (0, 1/2).$$

The source paper constructs smooth increasing functions g_m on $\mathbb{R}$, with $g_m = 0$ to the left of 0 and $g_m = 1$ to the right of 1, by the self-similar recursion

$$g_{m+1}(t) = \frac{1}{2}g_m(t/\rho) + \frac{1}{2}g_m\left(1 - \frac{1-t}{\rho}\right).$$

Equivalently, if $\mathcal{I}_m$ is the family of the 2^m basic intervals in the m th step of the symmetric Cantor construction with ratio ρ , and a_I is the left endpoint of I , then

$$g_m(t) = 2^{-m} \sum_{I \in \mathcal{I}_m} g_0\left(\frac{t - a_I}{\rho^m}\right).$$

For $N = 1$ set

$$\phi_I(t) = g_0 \left(\frac{t - a_I}{\rho^m} \right).$$

For $N > 1$, use the same tensor cutoff as the source paper: choose a fixed $\eta \in C_c^\infty(\mathbb{R}^N)$, equal to one on the relevant unit box, and set

$$\Phi_I(x) = 16 \phi_I(x_1) \eta(x).$$

For $N = 1$ this formula is read with $\eta \equiv 1$ and the harmless factor 16 omitted. Then the source functions used in the lower-bound construction have the form

$$u_m = 2^{-m} \sum_{I \in \mathcal{I}_m} \Phi_I.$$

Lemma 1. *There is a constant $A = A(N, \gamma, g_0, \eta)$ such that*

$$\mathfrak{q}_\gamma(\Phi_I) \leq A$$

for every level m and every $I \in \mathcal{I}_m$.

Proof. The L^1 gradient bound is uniform: the x_1 derivative has size ρ^{-m} on a strip of width ρ^m , while all cutoff-derivative terms are supported in a fixed compact set and are uniformly bounded.

For the weak endpoint term, write $r = \rho^m$. A single atom has only one transition layer of width comparable to r . Split pairs (x, y) into $|x - y| \leq r$ and $|x - y| > r$.

For $|x - y| \leq r$, Lipschitz control gives

$$|\Phi_I(x) - \Phi_I(y)| \lesssim r^{-1} |x - y|,$$

so membership in $E_{\lambda, \gamma}[\Phi_I]$ implies

$$|x - y| \gtrsim (\lambda r)^{1/|\gamma|}.$$

The transition layer has measure $O(r)$ in the normal direction, and integrating the kernel $|x - y|^{\gamma - N}$ over the admissible annuli gives

$$\nu_\gamma(E_{\lambda, \gamma}[\Phi_I] \cap \{|x - y| \leq r\}) \lesssim \lambda^{-1}.$$

For $|x - y| > r$, the atom behaves like a bounded jump across one Lipschitz layer. The set of points within distance h of that layer has measure $O(h)$ in the normal direction; integrating $h^\gamma dh$ up to the largest scale allowed by

$$1 \gtrsim \lambda h^{1+\gamma}$$

again gives a bound $\lesssim \lambda^{-1}$. The cutoff region contributes the same estimate with constants depending only on the fixed cutoff. Hence

$$\lambda \nu_\gamma(E_{\lambda, \gamma}[\Phi_I]) \leq C$$

uniformly in I, m, λ , proving the lemma. $\square$

On the other hand, the lower-bound construction in Section 6 of the source paper proves that these multiscale averages satisfy

$$\nu_\gamma(E_{1, \gamma}[u_m]) \geq c_{N, \gamma} m - C_{N, \gamma},$$

after the same tensor cutoff in higher dimensions. Therefore

$$\mathfrak{q}_\gamma(u_m) \geq c_{N,\gamma}m - C_{N,\gamma}.$$

Now normalize the atoms by A : each $A^{-1}\Phi_I$ belongs to the $\mathfrak{q}_\gamma$ -unit ball, while

$$A^{-1}u_m = 2^{-m} \sum_{I \in \mathcal{I}_m} A^{-1}\Phi_I$$

belongs to the convex hull of that unit ball. But

$$\mathfrak{q}_\gamma(A^{-1}u_m) = A^{-1}\mathfrak{q}_\gamma(u_m) \rightarrow \infty.$$

Thus the convex hull of the $\mathfrak{q}_\gamma$ -unit ball is not $\mathfrak{q}_\gamma$ -bounded. By the criterion at the start of the proof, $\dot{W}^{1,1}(\gamma)$ is not normable for $-1 < \gamma < 0$.

If $B\dot{V}(\gamma)$ were normable, then its linear subspace $\dot{W}^{1,1}(\gamma)$ with the inherited topology would be normable, contradicting what we just proved. Hence $B\dot{V}(\gamma)$ is not normable either. $\square$

Scope and Limitations

This proves a negative answer to the source paper’s normability question throughout the open interval $-1 < \gamma < 0$ and for every dimension $N \geq 1$.

The endpoint $\gamma = -1$ is not settled here. The reason is structural: the uniform one-scale atom bound used above becomes logarithmically unstable at $\gamma = -1$, exactly the endpoint where the source paper separates special cases in its generic-failure section.

Novelty Check

Local run indexes were searched for arXiv id 2109.02930, the title, “Sobolev norms”, “normable”, and close variants; no duplicate packet for this normability question was found.

On 2026-07-03, bounded web searches were run for phrases including “Families of functionals representing Sobolev norms normable gamma”, “dot $\dot{W}^{1,1}(\gamma)$ normable”, and “Are these spaces normable gamma Sobolev Brezis”. The searches surfaced the source arXiv page but no later explicit answer to the normability question. Novelty confidence is moderate, subject to expert review of the one-scale atom estimate.

Human Review Recommendation

Review as a likely valid partial solution. The key point to check is the uniform atom estimate in the lemma, especially the cutoff contribution in dimensions $N > 1$. The convex-hull obstruction is then immediate and standard.

References

- [1] Haim Brezis, Andreas Seeger, Jean Van Schaftingen, and Po-Lam Yung, *Families of functionals representing Sobolev norms*, arXiv:2109.02930, 2021.